



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SAI FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

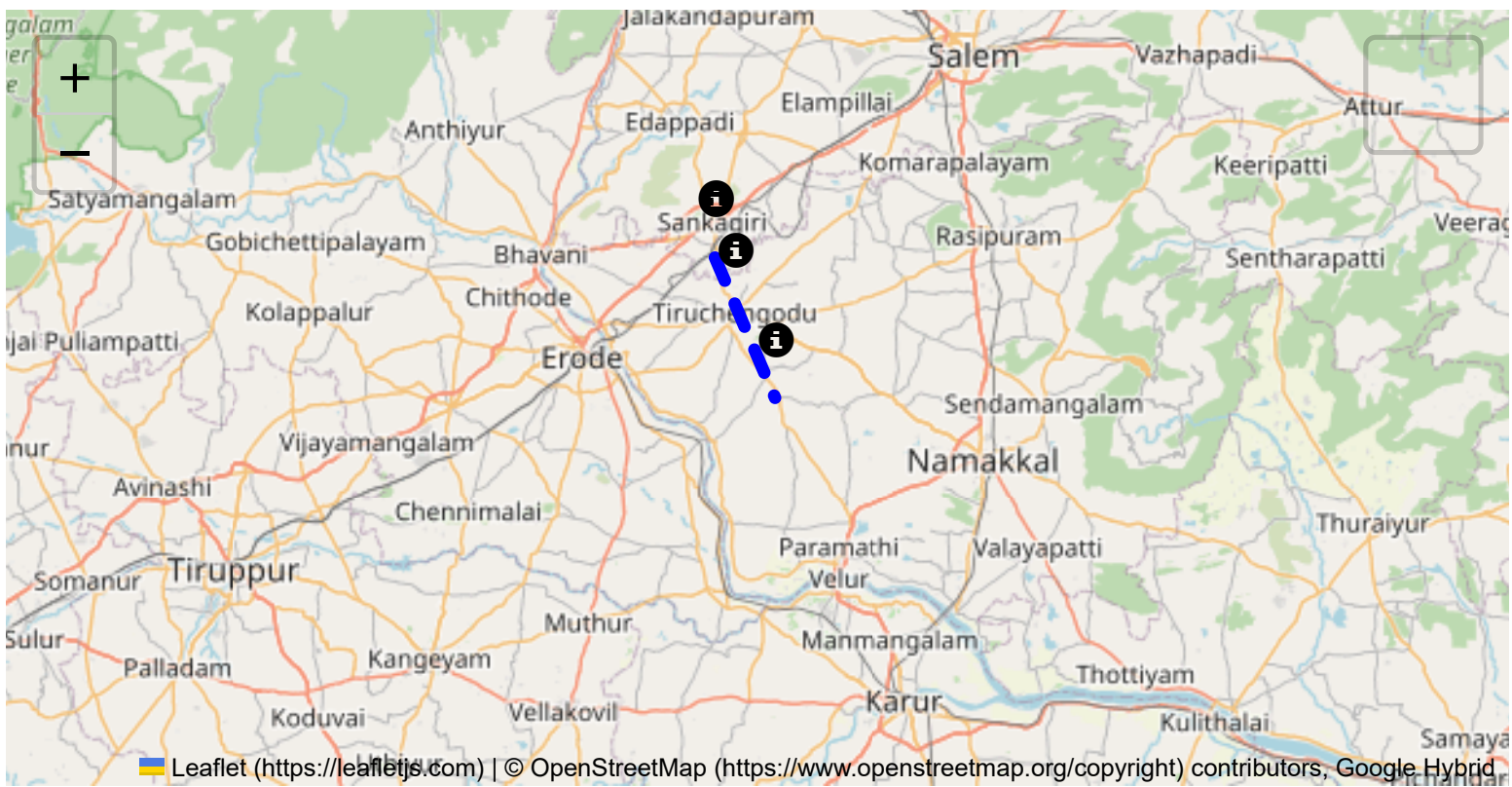
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

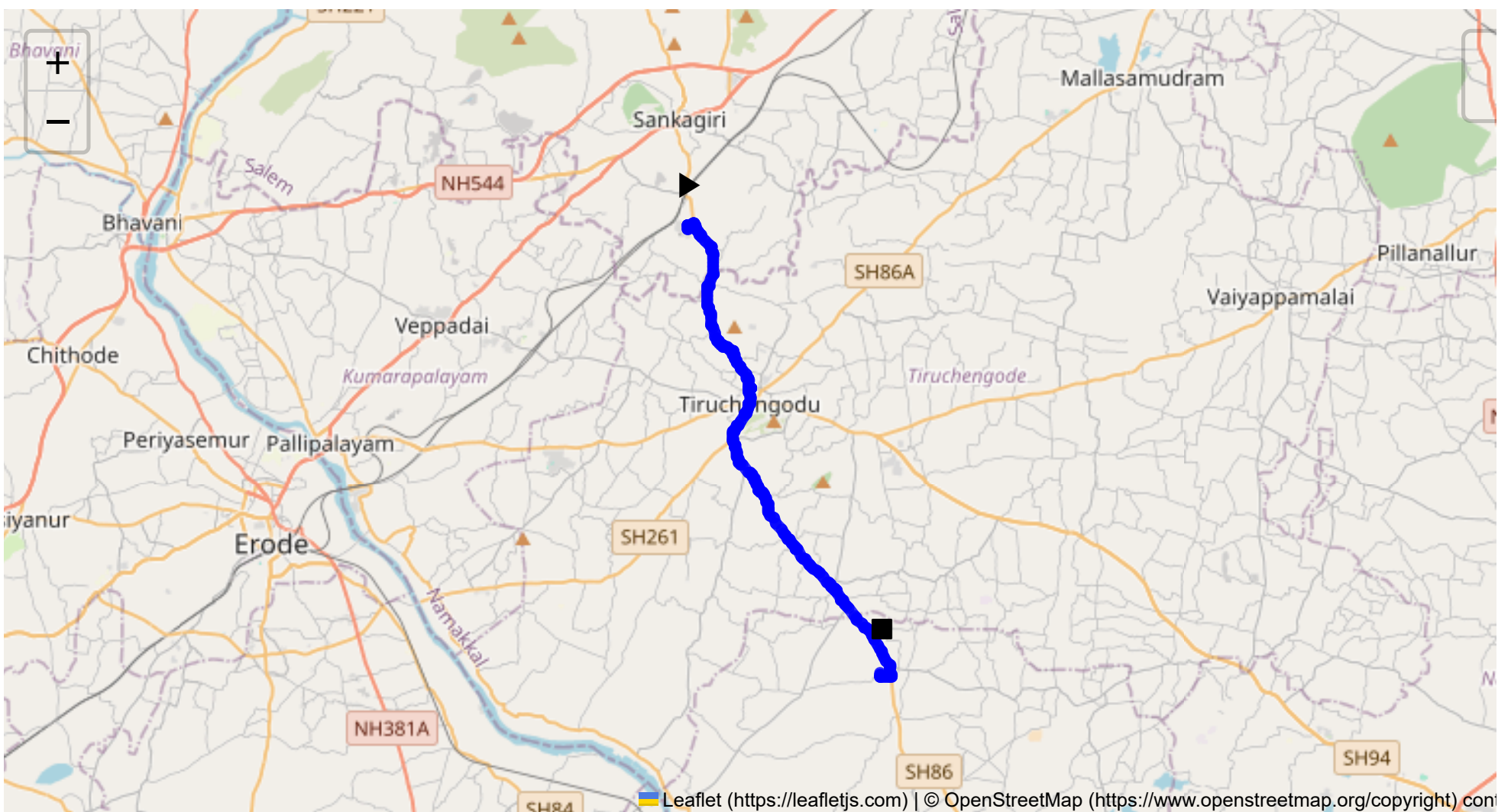
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 21.39 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.2860199, 77.9404457)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 7WPR+C5, Seyyampalayam, via Puthur and Tiruchengode, covers approximately 21.39 kilometers. It begins in Sangagiri, transitions through Puthur, and passes Tiruchengode via the central bus stand before reaching Seyyampalayam. The journey typically involves regional roads with varying traffic densities.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate. Monsoon season (June to November) can bring heavy rainfall, potentially causing localized flooding and slippery road conditions. Drivers should be cautious of water logging, reduced visibility, and hydroplaning during these times.

3. Analysis of Traffic Patterns

- **Peak Hours:** Generally, the morning (8 AM - 10 AM) and evening (5 PM - 7 PM) see increased traffic volumes, especially near Tiruchengode and the bus stand.
- **Congestion-Prone Areas:** The Tiruchengode Central Bus Stand is a major congestion point, especially during peak transit hours. Puthur also experiences moderate congestion during school and work commutes.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The roads primarily consist of single-lane or bi-lane surfaces, which may exhibit wear and tear. Maintenance quality is variable, with some sections having potholes or uneven surfaces.
- **Infrastructure:** There are limited pedestrian crossings and a few underpasses, but signages could be lacking in certain stretches.

5. Suggestions for Alternative Routes for Emergencies

In the event of road blockages, alternative routes could involve detouring via Periya Semur to the north or Kodumudi to the south, though these could extend travel time significantly due to longer distances.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Heavy vehicles carrying hazardous materials may face restrictions in town centers and near school zones during peak hours.
- It is essential to consult local transport officials for specific permissions or restrictions related to hazardous materials.

7. Overview of Historical Incidents

There are records of occasional accidents involving heavy vehicles on this route, typically within or near Tiruchengode, often attributed to road conditions or congestion. Incident rates are lower than on major highways.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** The region includes agricultural lands and some areas with biodiversity importance around Tiruchengode. Drivers should be aware of local wildlife crossings and community zones.

- **Environmental Impact:** Proper containment and spill-response plans are crucial to minimize ecological disturbances.

9. Analysis of Communication Coverage

Mobile network coverage along this route is generally reliable, with potential coverage gaps in more rural stretches around Puthur. Drivers should ensure that communication devices are equipped with offline maps and emergency contact numbers.

10. Estimated Emergency Response Times

- **Urban sections:** In Tiruchengode, response times are typically around 15 to 25 minutes due to proximity to public services.
- **Rural sections:** In areas like Puthur, response could range between 30 to 45 minutes due to fewer facilities and longer distances.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Seyyampalayam poses moderate risks chiefly due to road quality, weather conditions during the monsoon, and congestion around Tiruchengode. It is recommended that drivers plan travel times to avoid peak hours and monsoon period challenges. Maintaining constant communication, adhering to local regulations, and using updated navigation tools are essential for safe transit of hazardous materials. Routine vehicle checks and emergency preparedness, especially spill response and first aid readiness, are crucial for drivers on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.38386, 77.89482	15 KM/Hr
2	U-Turn	High	11.3839218, 77.8950042	10 KM/Hr
3	Turn	High	11.43811, 77.87348	15 KM/Hr
4	Turn	Medium	11.43961, 77.87341	30 KM/Hr
5	Turn	Medium	11.43971, 77.87348	30 KM/Hr
6	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
7	Turn	High	11.37868, 77.89498	15 KM/Hr
8	Turn	High	11.37850, 77.89453	15 KM/Hr
9	Turn	Medium	11.36439, 77.88943	30 KM/Hr
10	Turn	Medium	11.36434, 77.88942	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Blind Spot	Blind Spot	11.28479, 77.94438	10 KM/Hr
12	Turn	High	11.28476, 77.94029	15 KM/Hr
13	Turn	Medium	11.28571, 77.94026	30 KM/Hr
14	Turn	Medium	11.28579, 77.94021	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.38386, 77.89482



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.3839218, 77.8950042



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36439, 77.88943



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36434, 77.88942



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.28479, 77.94438



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.28476, 77.94029
